

5 Series B MSO

Mixed Signal Oscilloscope Datasheet



Key performance specifications

With a remarkably innovative pinch-swipe-zoom touchscreen user interface, the industry's largest high-definition display, and 4, 6, or 8 FlexChannel® inputs that let you measure one analog or eight digital signals per channel, the 5 Series MSO is ready for today's toughest challenges, and tomorrow's too. It sets a new standard for performance, analysis, and overall user experience.

Input channels

- 4, 6, or 8 FlexChannel® inputs
- Each FlexChannel provides:
 - One analog signal that can be displayed as a waveform view, a spectral view, or both simultaneously
 - Eight digital logic inputs with TLP058 logic probe

Bandwidth (all analog channels)

- 350 MHz, 500 MHz, 1 GHz, 2 GHz (upgradable)

Sample rate (all analog / digital channels)

- Real-time: 6.25 GS/s
- Interpolated: 500 GS/s

Record length (all analog / digital channels)

- 62.5 Mpoints standard
- 125, 250, 500 Mpoints, (optional and upgradable)

Waveform capture rate

- >500,000 waveforms/s

Vertical resolution

- 12-bit ADC
- Up to 16-bits in High Res mode

Standard trigger types

- Edge, Pulse Width, Runt, Timeout, Window, Logic, Setup & Hold, Rise/Fall Time, Parallel Bus, Sequence, Visual Trigger, Video (optional), RF vs. Time (optional)
- Auxiliary Trigger ≤5 VRMS, 50Ω, 250 MHz (>200 mV_{pp}) (Edge Trigger only)

Standard analysis

- Cursors: Waveform, V Bars, H Bars, V&H Bars
- Measurements: 36
- Spectrum View: Frequency-domain analysis with independent controls for frequency and time domains
- FastFrame™: Segmented memory acquisition mode with maximum trigger rate >5,000,000 waveforms per second
- Plots: Time Trend, Histogram, Spectrum and Phase Noise
- Math: Basic waveform arithmetic, FFT, and advanced equation editor
- Search: Search on any trigger criteria
- Jitter: TIE and Phase Noise

Optional analysis (optional and upgradable)

- Advanced Jitter and Eye Diagram Analysis
- User-defined filtering
- Advanced Spectrum View
- RF vs. Time traces, triggers, Spectrograms, and IQ capture
- Digital Power Management
- Mask/Limit Testing
- Inverters, Motors, and Drives
- LVDS Debug and Analysis
- PAM3 Analysis
- Advanced Power Measurements and Analysis
- Advanced Vector Signal Analysis (SignalVu-PC)

Optional protocol trigger, decode, and analysis (optional and upgradable)

I²C, SPI, eSPI, I3C, RS-232/422/485/UART, SPMI, SMBus, CAN, CAN FD, CAN XL, LIN, FlexRay, SENT, PSi5, CXPI, Automotive Ethernet, MIPI C-PHY, MIPI D-PHY, USB 2.0, eUSB2.0, Ethernet, EtherCAT, Audio, MIL-STD-1553, ARINC 429, Spacewire, 8B/10B, NRZ, Manchester, SVID, SDLC, 1-Wire, MDIO, and NFC